

Alex Chen, M.S.

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Professional Summary

Senior Machine Learning Engineer with 6+ years of experience specializing in distributed LLM serving, RAG architectures, and NLP pipeline optimization. Proven track record of scaling vector search systems to 50,000 QPS with sub-50ms latency at high-growth FinTech and cloud data platforms.

Technical Competencies

Languages: Python, Go, C++, SQL, TypeScript

ML & NLP Frameworks: PyTorch, HuggingFace Transformers, vLLM, Ray Serve, ONNX Runtime

Vector Databases & Search: ChromaDB, Qdrant, Milvus, Elasticsearch (BM25), FAISS

Infrastructure: Docker, Kubernetes, AWS (SageMaker, EKS), Terraform, Triton Inference Server

Professional Experience

Senior Machine Learning Engineer — Stripe, San Francisco, CA

March 2022 – Present

- Architected an enterprise-grade RAG dispute intelligence system indexing 12M+ financial dispute cases, reducing human review time by 38%.
- Optimized sentence-transformer embedding inference using TensorRT and vLLM on NVIDIA A100 GPUs, achieving a 4.2x throughput increase and saving \$240,000 in annual AWS compute costs.
- Deployed hybrid sparse-dense retrieval (BM25 + BGE-M3) with Reciprocal Rank Fusion, improving Top-5 retrieval recall from 78.4% to 92.1% on ambiguous fraud terminology.

Machine Learning Engineer — Databricks, Mountain View, CA

July 2019 – February 2022

- Built automated feature-store pipelines on Apache Spark supporting real-time model scoring for 40+ tenant applications.
- Co-authored the Lakehouse Vector Search connector, integrating HNSW approximate nearest-neighbor indexing into Delta Lake tables.

Education

Stanford University — M.S. in Computer Science (Artificial Intelligence Track)

September 2017 – June 2019 • GPA: 3.92/4.0

University of Washington — B.S. in Computer Engineering (Magna Cum Laude)

September 2013 – June 2017